1. Write code to display user input from EditText into TextView

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"

xmlns:tools="http://schemas.android.com/tools"
xmlns:app="http://schemas.android.com/apk/res
-auto"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">
  <EditText
     android:id="@+id/editTextInput"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:hint="Enter text"
     android:inputType="text"
    />
  <Button
     android:id="@+id/buttonShow"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Show" />
  <TextView
     android:id="@+id/textViewOutput"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Output" />
  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Executed by Your Name" />
</LinearLayout>
```

MainActivity.java:

```
package com.example.exp4_1;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
public class MainActivity extends
AppCompatActivity {
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
     super.onCreate(savedInstanceState);
     EdgeToEdge.enable(this);
     setContentView(R.layout.activity_main);
ViewCompat.setOnApplyWindowInsetsListener(f
indViewById(R.id.main), (v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.syst
emBars());
        v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
     });
     EditText editText =
findViewById(R.id.editTextInput);
     TextView textView =
findViewById(R.id.textViewOutput);
     Button button =
findViewById(R.id.buttonShow);
     button.setOnClickListener(v ->
textView.setText(editText.getText().toString()));
  }
}
```

2.Modify the app to show a greeting like “Hello, [user name]!” using input from EditText

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
xmlns:app="http://schemas.android.com/apk/res
-auto"
xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:background="#23aab4"
  tools:context=".MainActivity">
  <EditText
     android:id="@+id/editTextName"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:hint="Enter name"
     android:inputType="textPersonName"
     android:importantForAutofill="no" />
  <Button
     android:id="@+id/buttonGreet"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Greet" />
  <TextView
     android:id="@+id/textViewGreeting"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Greeting" />
  <!-- Simple Footer -->
  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="vesp Student”/>
</LinearLayout>
```

MainActivity.java:

```
package com.example.exp4_2;


import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;
public class MainActivity extends
AppCompatActivity {
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
     super.onCreate(savedInstanceState);
     EdgeToEdge.enable(this);
     setContentView(R.layout.activity_main);

ViewCompat.setOnApplyWindowInsetsListener(f
indViewById(R.id.main), (v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.syst
emBars());
        v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
     });

     EditText editText =
findViewById(R.id.editTextName);
     TextView textView =
findViewById(R.id.textViewGreeting);
     Button button =
findViewById(R.id.buttonGreet);

     button.setOnClickListener(v ->
          textView.setText("Hello, " +
editText.getText().toString() + "!"));


  }
}
```

3.Change the font size and color of the TextView using XML.

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
xmlns:app="http://schemas.android.com/apk/res
-auto"
  xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   tools:context=".MainActivity">
   <EditText
      android:id="@+id/editTextName"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter name"
      android:inputType="textPersonName"
      android:importantForAutofill="no" />
   <Button
      android:id="@+id/buttonGreet"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Greet" />
   <TextView
      android:id="@+id/textViewGreeting"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Greeting"
      android:textSize="34sp"
      android:textColor="#1E88E5"
      android:textStyle="bold" />
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="vesp studentf" />
</LinearLayout>
```

MainActivity.java:

```
package com.example.exp4_3;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;


import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;
public class MainActivity extends
AppCompatActivity {
   @Override
   protected void onCreate(Bundle
savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
ViewCompat.setOnApplyWindowInsetsListener(f
indViewById(R.id.main), (v, insets) -> {
         Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.syst
emBars());
         v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
         return insets;
      });
      EditText editText =
findViewById(R.id.editTextName);
      TextView textView =
findViewById(R.id.textViewGreeting);
      Button button =
findViewById(R.id.buttonGreet);
      button.setOnClickListener(v ->
            textView.setText("Hello, " +
editText.getText().toString() + "!"));


   }
}
```

4.Write code to display a message when a button is clicked.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"

xmlns:app="http://schemas.android.com/apk/res
-auto"

xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">

  <Button
     android:id="@+id/simpleButton"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Click Me" />
  <TextView
     android:id="@+id/footerText"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Made by your name"
     android:textSize="14sp"
     android:layout_marginTop="30dp"
     android:gravity="center" />

</LinearLayout>
```

MainActivity.java

```
package com.example.exp5_1;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;


public class MainActivity extends
AppCompatActivity {
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
     super.onCreate(savedInstanceState);
     EdgeToEdge.enable(this);
     setContentView(R.layout.activity_main);

ViewCompat.setOnApplyWindowInsetsListener(f
indViewById(R.id.main), (v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.syst
emBars());
        v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
     });
     Button simpleButton =
findViewById(R.id.simpleButton);
     simpleButton.setOnClickListener(new
View.OnClickListener() {
        @Override
        public void onClick(View v) {
           Toast.makeText(MainActivity.this, "This
button was clicked!!",
Toast.LENGTH_LONG).show();
        }
     });

  }
}
```

5.Change TextView Based on Toggle.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="ON"
        android:textOff="OFF"/>

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Toggle Status"
        android:textSize="22sp"
        android:padding="10dp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.program2;

import android.os.Bundle;
import android.widget.TextView;
import android.widget.ToggleButton;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        ToggleButton toggleButton =
findViewById(R.id.toggleButton);
        TextView textView =
findViewById(R.id.textView);


toggleButton.setOnCheckedChangeListener((bu
ttonView, isChecked) -> {
            if (isChecked) {
                textView.setText("Toggle is ON");
            } else {
                textView.setText("Toggle is OFF");
            }
        });
    }
}
```

6.Write and Execute code to display selected hobbies using CheckBox.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/a
ndroid"
xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   tools:context=".MainActivity">
   <CheckBox
      android:id="@+id/cbCricket"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Cricket" />
   <CheckBox
      android:id="@+id/cbMusic"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Music" />
   <CheckBox
      android:id="@+id/cbReading"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Reading" />
   <Button
      android:id="@+id/btnShow"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Show hobbies"
      android:onClick="showHobbies" />
   <TextView
      android:id="@+id/tvResult"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Selected hobbies will appear here"
      android:textSize="18sp"
      android:paddingTop="16dp" />
   <TextView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="done by your name"
      android:textSize="18sp"
      android:paddingTop="16dp" />
</LinearLayout>
```

MainActivity.java

```
package com.example.exp6_1;
import android.os.Bundle;
import android.view.View;
import android.widget.CheckBox;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
   CheckBox cbCricket, cbMusic, cbReading;
   TextView tvResult;
   @Override
   protected void onCreate(Bundle
savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);

ViewCompat.setOnApplyWindowInsetsListener(findVi
ewById(R.id.main), (v, insets) -> {
         Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemB
ars());
         v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
         return insets;   });
      cbCricket = findViewById(R.id.cbCricket);
      cbMusic   = findViewById(R.id.cbMusic);
      cbReading = findViewById(R.id.cbReading);
      tvResult  = findViewById(R.id.tvResult);
   }
   public void showHobbies(View v) {
      String s = "";
      if (cbCricket.isChecked()) s += "Cricket ";
      if (cbMusic.isChecked())   s += "Music ";
      if (cbReading.isChecked()) s += "Reading ";
      if (s.isEmpty()) s = "No hobby selected";
      tvResult.setText(s);
   }
}
```

7.Write and Execute code to clear all CheckBox and RadioButton selections on button click.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"

xmlns:app="http://schemas.android.com/apk/res
-auto"
  xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   tools:context=".MainActivity">
   <CheckBox
      android:id="@+id/cbCricket"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Cricket" />
   <CheckBox
      android:id="@+id/cbMusic"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Music" />
   <CheckBox
      android:id="@+id/cbReading"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Reading" />
   <Button
      android:id="@+id/btnShow"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Show hobbies"
      android:onClick="showHobbies" />
   <TextView
      android:id="@+id/tvResult"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Selected hobbies will appear
here"
      android:textSize="18sp"
      android:paddingTop="8dp" />
   <Button
      android:id="@+id/btnCount"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Count selected"
      android:onClick="countHobbies"
      android:layout_marginTop="16dp" />
   <TextView
      android:id="@+id/tvCount"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Count will appear here"
      android:textSize="18sp"
      android:paddingTop="4dp" />
   <RadioGroup
      android:id="@+id/radioGroup"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:orientation="horizontal"
      android:layout_marginTop="24dp">
      <RadioButton
         android:id="@+id/rbMale"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Male" />
      <RadioButton
         android:id="@+id/rbFemale"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Female" />
   </RadioGroup>
   <Button
      android:id="@+id/btnClear"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Clear all"
      android:onClick="clearAll"
      android:layout_marginTop="16dp" />
   <TextView
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="done by your name"
      android:textSize="18sp"
      android:paddingTop="4dp" />
</LinearLayout>
```

MainActivity.Java file

```
package com.example.exp6_3;

import android.os.Bundle;
import android.view.View;
import android.widget.CheckBox;
import android.widget.RadioGroup;
import android.widget.TextView;


import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;
public class MainActivity extends
AppCompatActivity {

  CheckBox cbCricket, cbMusic, cbReading;
  TextView tvResult, tvCount;
  RadioGroup radioGroup;
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);

ViewCompat.setOnApplyWindowInsetsListener(f
indViewById(R.id.main), (v, insets) -> {
          Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.syst
emBars());
          v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
          return insets;
      });
      cbCricket = findViewById(R.id.cbCricket);
      cbMusic   = findViewById(R.id.cbMusic);
      cbReading = findViewById(R.id.cbReading);
      tvResult  = findViewById(R.id.tvResult);
      tvCount   = findViewById(R.id.tvCount);
      radioGroup =
findViewById(R.id.radioGroup);
  }
  public void showHobbies(View v) {
      String s = "";
      if (cbCricket.isChecked()) s += "Cricket ";
      if (cbMusic.isChecked())   s += "Music ";
      if (cbReading.isChecked()) s += "Reading ";
      if (s.isEmpty()) s = "No hobby selected";
      tvResult.setText(s);
  }
  public void countHobbies(View v) {
      int count = 0;
      if (cbCricket.isChecked()) count++;
      if (cbMusic.isChecked())   count++;
      if (cbReading.isChecked()) count++;
      tvCount.setText("Selected: " + count);
  }
  public void clearAll(View v) {
      cbCricket.setChecked(false);
      cbMusic.setChecked(false);
      cbReading.setChecked(false);
      radioGroup.clearCheck();



  }
}
```

8.Write code to change the text color of a RadioButton when selected

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk
/res/android"

xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp"
  tools:context=".MainActivity">
  <RadioGroup
     android:id="@+id/radioGroup"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:orientation="horizontal">
     <RadioButton
        android:id="@+id/rbMale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male"
    android:textColor="@android:color/black" />
     <RadioButton
        android:id="@+id/rbFemale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female"
   android:textColor="@android:color/black" />
  </RadioGroup>
  <TextView
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:text="executed by kani"
          android:textSize="18sp"
     android:paddingTop="8dp" />
</LinearLayout>
```

MainActivity.Java file

```
package com.example.prac8;
import android.os.Bundle;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.graphics.Color;
import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import
androidx.core.view.WindowInsetsCompat;
public class MainActivity extends
AppCompatActivity {
  RadioGroup radioGroup;
  RadioButton rbMale, rbFemale;
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
     super.onCreate(savedInstanceState);
     EdgeToEdge.enable(this);
     setContentView(R.layout.activity_main);

ViewCompat.setOnApplyWindowInsetsListener(f
indViewById(R.id.main), (v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.syst
emBars());
        v.setPadding(systemBars.left,
systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
     });
     radioGroup =
findViewById(R.id.radioGroup);
     rbMale     = findViewById(R.id.rbMale);
     rbFemale   = findViewById(R.id.rbFemale);
radioGroup.setOnCheckedChangeListener(new
RadioGroup.OnCheckedChangeListener() {
        @Override
        public void
onCheckedChanged(RadioGroup group, int
checkedId) {
           rbMale.setTextColor(Color.BLACK);
           rbFemale.setTextColor(Color.BLACK);
           if (checkedId == R.id.rbMale) {
              rbMale.setTextColor(Color.GREEN);
} else if (checkedId == R.id.rbFemale) {
              rbFemale.setTextColor(Color.RED);
           }
        }
     }); }  }
```

9.Implement a circular indeterminate progress bar that appears during data loading.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <!-- Circular Indeterminate Progress Bar -->
    <ProgressBar
        android:id="@+id/circularProgress"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:indeterminate="true"
        android:visibility="gone"/>

    <!-- Button -->
    <Button
        android:id="@+id/btnLoad"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Load Data"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.prac9;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;
import
androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends
AppCompatActivity {
    ProgressBar circularProgress;
    Button btnLoad;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        circularProgress =
findViewById(R.id.circularProgress);
        btnLoad = findViewById(R.id.btnLoad);

        btnLoad.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {

                // Show loader

circularProgress.setVisibility(View.VISIBLE);

                // Simulate loading (3 sec)
                new
Handler(Looper.getMainLooper()).postDelayed(n
ew Runnable() {
                    @Override
                    public void run() {

circularProgress.setVisibility(View.GONE);
                    }
                }, 3000);
            }
        });
    }
}
```

10.Add gender selection using radio buttons and display the selected gender on login.

acitivty_main.xml code

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"

xmlns:app="http://schemas.android.com/apk/res
-auto"
  xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   tools:context=".MainActivity">
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Login Form"
      android:textSize="22sp"
      android:textStyle="bold"
      android:layout_marginBottom="20dp" />
   <EditText
      android:id="@+id/etUsername"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Username" />
   <EditText
      android:id="@+id/etPassword"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Password"
      android:inputType="textPassword"
      android:layout_marginTop="12dp" />
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Select Gender"
      android:layout_marginTop="16dp"
      android:textStyle="bold" />
   <RadioGroup
      android:id="@+id/radioGroupGender"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:orientation="horizontal">
<RadioButton
         android:id="@+id/rbMale"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Male" />
      <RadioButton
         android:id="@+id/rbFemale"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Female" />
   </RadioGroup>
   <Button
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Login"
      android:onClick="login"
      android:layout_marginTop="20dp" />
   <TextView
      android:id="@+id/tvResult"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text=""
      android:textSize="18sp"
      android:layout_marginTop="16dp" />
   <TextView
      android:id="@+id/tvResult"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Done by your name"
      android:textSize="18sp"
      android:layout_marginTop="16dp" />
</LinearLayout>
```

MainActivity.java code

```
package com.example.exp8_2;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;


import androidx.activity.EdgeToEdge;
import
androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends
AppCompatActivity {

   EditText etUsername, etPassword;
   RadioGroup radioGroupGender;
   TextView tvResult;
   @Override
   protected void onCreate(Bundle
savedInstanceState) {
       super.onCreate(savedInstanceState);
       EdgeToEdge.enable(this);
       setContentView(R.layout.activity_main);
       etUsername =
findViewById(R.id.etUsername);
       etPassword =
findViewById(R.id.etPassword);
       radioGroupGender =
findViewById(R.id.radioGroupGender);
       tvResult = findViewById(R.id.tvResult);
   }
   public void login(View v) {
       String username =
etUsername.getText().toString().trim();
       String password =
etPassword.getText().toString().trim();
       int selectedGenderId =
radioGroupGender.getCheckedRadioButtonId();
       if (username.isEmpty() ||
password.isEmpty()) {
           tvResult.setText("Username and
Password cannot be empty");
           return;
       }
       if (selectedGenderId == -1) {
           tvResult.setText("Please select gender");
           return;
       }
       RadioButton rb =
findViewById(selectedGenderId);
       String gender = rb.getText().toString();
       tvResult.setText("Login
Successful\nGender: " + gender);
   }
}
```

11.Design a registration form using LinearLayout that includes fields for Name, Email, Phone, and a Register button.

activity_main.xml code

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/a
ndroid"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="20dp">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Registration Form"
    android:textSize="22sp"
    android:textStyle="bold"
    android:layout_gravity="center"
    android:paddingBottom="20dp" />

  <EditText
    android:id="@+id/etName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Name"
    android:inputType="textPersonName" />

  <EditText
    android:id="@+id/etEmail"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Email"
    android:inputType="textEmailAddress"
    android:layout_marginTop="10dp" />

  <EditText
    android:id="@+id/etPhone"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Phone"
    android:inputType="phone"
    android:layout_marginTop="10dp" />
  <Button
    android:id="@+id/btnRegister"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Register"
    android:layout_marginTop="20dp" />

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Executed by kanishka"
    android:layout_marginTop="30dp"
    android:textSize="16sp" />

</LinearLayout>
```

MainActivity.java code

```
package com.example.prac11;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  EditText etName, etEmail, etPhone;
  Button btnRegister;

  @Override
  protected void onCreate(Bundle
savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    etName = findViewById(R.id.etName);
    etEmail = findViewById(R.id.etEmail);
    etPhone = findViewById(R.id.etPhone);
    btnRegister = findViewById(R.id.btnRegister);
    btnRegister.setOnClickListener(view -> {
      String name =
etName.getText().toString().trim();
      String email =
etEmail.getText().toString().trim();
      String phone =
etPhone.getText().toString().trim();

      if(name.isEmpty() || email.isEmpty() ||
phone.isEmpty()){
        Toast.makeText(MainActivity.this, "Please fill
all fields", Toast.LENGTH_SHORT).show();
      } else {
        Toast.makeText(MainActivity.this,
            "Name: " + name + "\nEmail: " + email
+ "\nPhone: " + phone,
            Toast.LENGTH_LONG).show();
      }
    });
  }
}
```

12.Include a CheckBox for "Terms and Conditions" and validate that it is checked before showing the toast.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <CheckBox
        android:id="@+id/cbTerms"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="I agree to Terms and
Conditions"/>

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Submit"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.exp9_2;

import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    CheckBox cbTerms;
    Button btnSubmit;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        cbTerms = findViewById(R.id.cbTerms);
        btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(v -> {

            if (cbTerms.isChecked()) {
                Toast.makeText(this, "Proceeding...",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Please accept
Terms and Conditions",
Toast.LENGTH_SHORT).show();
            }

        });
    }
}
```

13.Design a price summary table using TableLayout with item price, tax, shipping, and total.

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="20dp"
  android:background="#F5F5F5">

  <!-- Title -->
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Price Summary"
    android:textSize="22sp"
    android:textStyle="bold"
    android:textColor="#1A73E8"
    android:layout_gravity="center"
    android:paddingBottom="20dp"/>

  <!-- Table -->
  <TableLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:stretchColumns="1"
    android:background="#FFFFFF"
    android:padding="10dp">

    <TableRow>
      <TextView
        android:text="Item Price"
        android:textColor="#000000"
        android:padding="10dp"/>

      <TextView
        android:text="₹1000"
        android:textColor="#000000"
        android:padding="10dp"/>
    </TableRow>

    <TableRow>
      <TextView
        android:text="Tax"
        android:textColor="#000000"
        android:padding="10dp"/>

      <TextView
        android:text="₹100"
        android:textColor="#000000"
        android:padding="10dp"/>
    </TableRow>

    <TableRow>
      <TextView
        android:text="Shipping"
        android:textColor="#000000"
        android:padding="10dp"/>

      <TextView
        android:text="₹50"
        android:textColor="#000000"
        android:padding="10dp"/>
    </TableRow>

    <TableRow>
      <TextView
        android:text="Total"
        android:textStyle="bold"
        android:textColor="#1A73E8"
        android:padding="10dp"/>

      <TextView
        android:text="₹1150"
        android:textStyle="bold"
        android:textColor="#1A73E8"
        android:padding="10dp"/>
    </TableRow>

  </TableLayout>

  <!-- Footer -->
  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Performed by kanishka"
    android:textColor="#555555"
    android:layout_gravity="center"
```

```
    android:layout_marginTop="30dp"
    android:textSize="14sp"/>

</LinearLayout>
```

Optional

2.1 Create a new file inside /drawable as table_cell_border.xml and put the following code in there :

```
<?xml version="1.0" encoding="utf-8"?>
<shape
xmlns:android="http://schemas.android.com/apk
/res/android">

  <solid android:color="#FFFFFF"/>
  <stroke
    android:width="1dp"
    android:color="#000000"/>
  <padding
    android:left="6dp"
    android:top="6dp"
    android:right="6dp"
    android:bottom="6dp"/>

</shape>
```

(Use this to create cell borders for the table , otherwise skip it)

14.Create a product card using FrameLayout with an image and a discount label overlay.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLay
out
xmlns:android="http://schemas.android.com/apk
/res/android"
xmlns:app="http://schemas.android.com/apk/res
-auto"
xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity"
  android:orientation="vertical"
  >
  <FrameLayout
    android:id="@+id/productCard"
    android:layout_width="280dp"
    android:layout_height="200dp"
    android:background="#DDDDDD"
    android:padding="6dp"
    app:layout_constraintTop_toTopOf="parent"

app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="24dp">
    <ImageView
      android:id="@+id/productImage"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:src="@drawable/cat"
      android:scaleType="fitCenter" />
    <TextView
      android:id="@+id/discountLabel"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="20% OFF"
      android:textColor="@android:color/white"
      android:background="#E53935"
      android:padding="6dp"
      android:layout_gravity="top|end" />
  </FrameLayout>
  <TextView
  android:id="@+id/studentName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Performed By your name"
    android:textColor="@android:color/white"
    android:background="@android:color/black"
    android:padding="8dp"
app:layout_constraintBottom_toBottomOf="pare
nt"

app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginBottom="48dp" />
</androidx.constraintlayout.widget.ConstraintLay
out>
```

15.Use ImageView to show a banner image for a hotel welcome screen.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:background="#F5F5F5">

   <!-- Banner Image -->
   <ImageView
      android:id="@+id/bannerImage"
      android:layout_width="match_parent"
      android:layout_height="220dp"
      android:scaleType="centerCrop"
      android:src="@drawable/hotel_banner"
      android:contentDescription="Hotel
Banner"/>

   <!-- Welcome Text -->
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Welcome to Paradise Hotel"
      android:textSize="24sp"
      android:textStyle="bold"
      android:textColor="#333333"
      android:layout_marginTop="20dp"
      android:layout_gravity="center"/>

   <!-- Subtitle -->
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Enjoy Luxury & Comfort"
      android:textSize="16sp"
      android:textColor="#777777"
      android:layout_marginTop="8dp"
      android:layout_gravity="center"/>

   <!-- Button -->
   <Button
      android:id="@+id/btnBook"
      android:layout_width="200dp"
      android:layout_height="wrap_content"
      android:text="Book Now"
      android:layout_marginTop="30dp"
      android:layout_gravity="center"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.hotelbanner;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

   @Override
   protected void onCreate(Bundle b) {
      super.onCreate(b);
      setContentView(R.layout.activity_main);

      Button btn = findViewById(R.id.btnBook);

      btn.setOnClickListener(v ->
            Toast.makeText(this, "Booking
Started!", Toast.LENGTH_SHORT).show()
      );
   }
}
```

16.Display a grid of book covers using GridView for a library catalog.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/a
ndroid"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="10dp">
  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Library Catalog"
     android:textSize="40sp"
     android:textStyle="bold"
     android:layout_marginBottom="10dp"/>
  <GridView
     android:id="@+id/gridView"
     android:layout_width="match_parent"
     android:layout_height="wrap_content"
     android:numColumns="2"
     android:horizontalSpacing="5dp"
     android:verticalSpacing="10dp"
     android:stretchMode="columnWidth"/>
  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="done by your name"
     android:textSize="20sp"
     android:textStyle="bold"
     android:layout_marginTop="20dp"
     android:layout_marginLeft="200dp"/>
</LinearLayout>
```

MainActivity.java file

```
package com.example.exp11_1;
import android.content.Context;
import android.os.Bundle;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.GridView;
import android.widget.ImageView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
  GridView gridView;
  int[] bookImages = {
        R.drawable.book1,
 R.drawable.book2,
        R.drawable.book3,
        R.drawable.book4
  };
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);
     gridView = findViewById(R.id.gridView);
  gridView.setAdapter(new BookAdapter(this));
  }
  class BookAdapter extends BaseAdapter {
     Context context;
     BookAdapter(Context context) {
        this.context = context;
 }
     @Override
     public int getCount() {
        return bookImages.length;
 }
     @Override
     public Object getItem(int position) {
        return bookImages[position];
}
     @Override
     public long getItemId(int position) {
        return position;
}
     @Override
     public View getView(int position, View
convertView, ViewGroup parent) {
        ImageView imageView;
        if (convertView == null) {
           imageView = new ImageView(context);
           imageView.setLayoutParams(
      new GridView.LayoutParams(500, 800));
imageView.setScaleType(ImageView.ScaleType.CEN
TER_CROP);
           imageView.setPadding(8, 8, 8, 8);
        } else {
           imageView = (ImageView) convertView;
        }
imageView.setImageResource(bookImages[position]);
        return imageView;
     }
  }
}
```

17. Design a calculator UI using GridLayout with buttons for digits and basic operators.(Add, subtract)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout
xmlns:android="http://schemas.android.com/apk/res/a
ndroid"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:columnCount="4"
   android:padding="16dp">
   <TextView
      android:id="@+id/tvDisplay"
      android:layout_width="match_parent"
      android:layout_height="80dp"
      android:layout_columnSpan="4"
      android:text="0"
      android:textSize="32sp"
      android:gravity="end|center_vertical"/>
   <Button android:id="@+id/btn1" android:text="1"/>
   <Button android:id="@+id/btn2" android:text="2"/>
   <Button android:id="@+id/btn3" android:text="3"/>
   <Button android:id="@+id/btnAdd"
android:text="+"/>
   <Button android:id="@+id/btn4" android:text="4"/>
   <Button android:id="@+id/btn5" android:text="5"/>
   <Button android:id="@+id/btn6" android:text="6"/>
   <Button android:id="@+id/btnSub"
android:text="-"/>
   <Button android:id="@+id/btn7" android:text="7"/>
   <Button android:id="@+id/btn8" android:text="8"/>
   <Button android:id="@+id/btn9" android:text="9"/>
   <Button android:id="@+id/btnEq"
android:text="="/>
   <Button android:id="@+id/btn0"
      android:layout_columnSpan="2"
      android:text="0"/>
   <Button android:id="@+id/btnClear"
      android:layout_columnSpan="2"
      android:text="CLEAR"/>
</GridLayout>
```

ActivityMain.java

```
package com.example.myapplication;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
   TextView tv;
   double num1 = 0;
  String operator = "";
   boolean isNewInput = true;
   @Override
   protected void onCreate(Bundle
savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      tv = findViewById(R.id.tvDisplay);
      int[] buttons = {
         R.id.btn0, R.id.btn1, R.id.btn2, R.id.btn3,
            R.id.btn4, R.id.btn5, R.id.btn6,
            R.id.btn7, R.id.btn8, R.id.btn9,
            R.id.btnAdd, R.id.btnSub,
            R.id.btnEq, R.id.btnClear
      };
      for (int id : buttons) {
         findViewById(id).setOnClickListener(v ->
handleClick((Button) v));
      }   }
   private void handleClick(Button button) {
      String value = button.getText().toString();
      if (value.matches("[0-9]")) {
         if (isNewInput) {
            tv.setText(value);
            isNewInput = false;
         } else {
            tv.append(value);
         }       }
      else if (value.equals("+") || value.equals("-")) {
         num1 =
Double.parseDouble(tv.getText().toString());
         operator = value;
         isNewInput = true;
      }
      else if (value.equals("=")) {
         double num2 =
Double.parseDouble(tv.getText().toString());
         double result = operator.equals("+") ? num1 +
num2 : num1 - num2;
         tv.setText(String.valueOf(result));
         isNewInput = true;
      }
      else if (value.equals("CLEAR")) {
         tv.setText("0");
         num1 = 0;
         operator = "";
         isNewInput = true;
      }
   }
}
```

18.Create a temperature converter from Celsius to Fahrenheit.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <EditText
        android:id="@+id/edit_input"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Celsius"
        android:inputType="numberDecimal"/>

    <Button
        android:id="@+id/button_convert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Convert"/>

    <TextView
        android:id="@+id/text_result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result"
        android:textSize="18sp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.exp14;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    EditText input;
    TextView result;
    Button convert;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        input = findViewById(R.id.edit_input);
        result = findViewById(R.id.text_result);
        convert =
findViewById(R.id.button_convert);

        convert.setOnClickListener(v -> {
            String value = input.getText().toString();

            if (!value.isEmpty()) {
                double celsius =
Double.parseDouble(value);
                double fahrenheit = (celsius * 9/5) +
32;

                result.setText("Fahrenheit: " +
fahrenheit);
            } else {
                result.setText("Enter value");
            }
        });
    }
}
```

19.Write a program in Android to implement a Date Picker using DatePickerDialog.

activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">
  <Button
     android:id="@+id/btnDate"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="Select Date"/>
  <TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:layout_gravity="center"
     android:layout_margin="12dp"
     android:text="Performed By your name"
     ></TextView>
</LinearLayout>
```

MainActivity.java code:

```
package com.example.exp16;
import android.app.DatePickerDialog;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import
androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends
AppCompatActivity {
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
     super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
     Button btn = findViewById(R.id.btnDate);
     btn.setOnClickListener(v -> {
        Calendar c = Calendar.getInstance();
        DatePickerDialog dp = new
DatePickerDialog(
              this,
              (view, y, m, d) ->
                    Toast.makeText(this,
                          d + "/" + (m + 1) + "/" + y,

Toast.LENGTH_SHORT).show(),
              c.get(Calendar.YEAR),
              c.get(Calendar.MONTH),
              c.get(Calendar.DAY_OF_MONTH)
        );
        // Restrict past dates

dp.getDatePicker().setMinDate(System.currentTi
meMillis());
        dp.show();
     });
  }
}
```

20.Write a program to create two activities – Login Activity and Welcome Activity.

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Email Address"
        android:inputType="textEmailAddress" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:hint="Password"
        android:inputType="textPassword" />

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Login" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="30dp"
        android:text="Executed by Raj Sangle"
        android:textStyle="bold"
        android:textSize="16sp" />

</LinearLayout>
```

MainActivity.java:

```
package com.example.mavajava; // Change to
your package name

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    EditText etEmail, etPassword;
    Button btnLogin;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etEmail = findViewById(R.id.etEmail);
        etPassword =
findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String email =
etEmail.getText().toString();
                String pass =
etPassword.getText().toString();

                if (email.isEmpty() || pass.isEmpty()) {
                    Toast.makeText(MainActivity.this,
"Please enter all details",
Toast.LENGTH_SHORT).show();
                } else if
(email.equals("rhn@gmail.com") &&
pass.equals("556062")) {
```

```
                Intent intent = new
Intent(MainActivity.this, WelcomeActivity.class);
                startActivity(intent);
                Toast.makeText(MainActivity.this,
"Login Successful",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(MainActivity.this,
"Invalid Credentials",
Toast.LENGTH_SHORT).show();
            }
        }
    });
  }
}
```

activity_welcome.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome!"
        android:textSize="30sp"
        android:textStyle="bold" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Login Successful."
        android:textSize="18sp" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:text="Executed by Raj Sangle"
        android:textColor="#555" />

</LinearLayout>
```

WelcomeActivity.java:

```
package com.example.mavajava;



import android.os.Bundle;
import
androidx.appcompat.app.AppCompatActivity;
public class WelcomeActivity extends
AppCompatActivity {
    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_welcome);
    }
}
```

AndroidManifest.xml:

```
<manifest
xmlns:android="http://schemas.android.com/apk
/res/android"
xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
      android:theme="@style/Theme.MavaJava">
        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action
android:name="android.intent.action.MAIN" />
                <category
android:name="android.intent.category.LAUNCH
ER" />
            </intent-filter>
        </activity>
     <activity android:name=".WelcomeActivity" />
    </application>
</manifest>
```

21.What code is used to open a website (e.g., “https://www.google.com”) using an implicit intent?

activity_main.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/btnOpen"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Website"/>

</LinearLayout>
```

MainActivity.java

```
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    Button btnOpen;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnOpen = findViewById(R.id.btnOpen);

        btnOpen.setOnClickListener(v -> {
            Intent intent = new
Intent(Intent.ACTION_VIEW);
intent.setData(Uri.parse("https://www.google.co
m"));
            startActivity(intent);
        });
    }
}
```

22.Write code to check the current Wi-Fi connection status.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <Button
      android:id="@+id/btnWifiStatus"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Check WiFi Status" />
   <TextView
      android:id="@+id/textOutput"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Status"
      android:textSize="18sp"
      android:padding="10dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.expp;

import android.net.wifi.WifiManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import
androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends
AppCompatActivity {
  WifiManager wifiManager;
   TextView textOutput;
  @Override
   protected void onCreate(Bundle
savedInstanceState) {
      super.onCreate(savedInstanceState);

   setContentView(R.layout.activity_main);
      wifiManager = (WifiManager)
getApplicationContext().getSystemService(WIFI
_SERVICE);
      Button btnWifiStatus =
findViewById(R.id.btnWifiStatus);
      textOutput = findViewById(R.id.textOutput);
      btnWifiStatus.setOnClickListener(v -> {
         if (wifiManager.isWifiEnabled()) {
            textOutput.setText("WiFi is ON");
         } else {
            textOutput.setText("WiFi is OFF");
         }
      });
   }
}
```

Manifest File

```
<?xml version="1.0" encoding="utf-8"?>
<manifest
xmlns:android="http://schemas.android.com/apk
/res/android">
   <uses-permission
android:name="android.permission.ACCESS_W
IFI_STATE"/>
   <application
      android:allowBackup="true"
      android:label="WiFiStatusApp"
      android:supportsRtl="true"
android:theme="@style/Theme.AppCompat.Ligh
t.DarkActionBar">
      <activity
         android:name=".MainActivity"
         android:exported="true">
         <intent-filter>
            <action
android:name="android.intent.action.MAIN" />
            <category
android:name="android.intent.category.LAUNCH
ER" />
         </intent-filter>
      </activity>
   </application>
</manifest>
```

23.Write a code snippet to listen for Airplane Mode state changes.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/a
ndroid"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Airplane Mode Listener"
        android:textSize="20sp"/>
</LinearLayout>
```

Manifest file

```
<?xml version="1.0" encoding="utf-8"?>
<manifest
xmlns:android="http://schemas.android.com/apk/res/a
ndroid">
    <application
        android:allowBackup="true"
        android:label="AirplaneModeApp"
        android:supportsRtl="true"
android:theme="@style/Theme.AppCompat.Light.Dar
kActionBar">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action
android:name="android.intent.action.MAIN" />
                <category
android:name="android.intent.category.LAUNCHER"
/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

MainActivity.java

```
package com.example.exp21;
import android.content.IntentFilter;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
 ModeChangeReceiver receiver;
    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        receiver = new ModeChangeReceiver();
    }
    @Override
    protected void onResume() {
        super.onResume();
        IntentFilter filter = new IntentFilter();
filter.addAction("android.intent.action.AIRPLANE_MO
DE");
        registerReceiver(receiver, filter);
    }
    @Override
    protected void onPause() {
        super.onPause();
        unregisterReceiver(receiver);
    }
}
```

ModeChangeReceiver.java

```
package com.example.exp21;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.provider.Settings;
import android.widget.Toast;
public class ModeChangeReceiver extends
BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent
intent) {
        if
(intent.getAction().equals(Intent.ACTION_AIRPLANE
_MODE_CHANGED)) {
            boolean isOn = Settings.Global.getInt(
                    context.getContentResolver(),
Settings.Global.AIRPLANE_MODE_ON, 0) != 0;
            if (isOn) {
                Toast.makeText(context, "Airplane Mode
ON", Toast.LENGTH_LONG).show();
            } else {
                Toast.makeText(context, "Airplane Mode
OFF", Toast.LENGTH_LONG).show();
            }
        }
    }}
```

24.Write a program to send email.(using intent)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Email" />

</LinearLayout>
```

MainActivity.java

```
package com.example.exp27;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        Button send = findViewById(R.id.btnSend);

        send.setOnClickListener(v -> {
            Intent i = new
Intent(Intent.ACTION_SEND);
            i.setType("message/rfc822");
            i.putExtra(Intent.EXTRA_EMAIL, new
String[]{"test@gmail.com"});
            i.putExtra(Intent.EXTRA_SUBJECT,
"Test Mail");
            i.putExtra(Intent.EXTRA_TEXT, "Hello
from Android");
            startActivity(Intent.createChooser(i,
"Send Email"));
        });
    }
}
```

25.Check flashlight status of phone using simple program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk
/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnCheck"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Check Flashlight Status"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.flashlightstatus;

import android.content.Context;
import
android.hardware.camera2.CameraManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    CameraManager cameraManager;
    String cameraId;
    boolean isTorchOn = false;

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);
        Button btnCheck =
findViewById(R.id.btnCheck);

        cameraManager = (CameraManager)
getSystemService(Context.CAMERA_SERVICE
);

        try {
            cameraId =
cameraManager.getCameraIdList()[0];


cameraManager.registerTorchCallback(new
CameraManager.TorchCallback() {
                @Override
                public void
onTorchModeChanged(String id, boolean
enabled) {
                    isTorchOn = enabled;
                }
            }, null);

        } catch (Exception e) {
            e.printStackTrace();
        }

        btnCheck.setOnClickListener(v -> {
            if (isTorchOn) {
                Toast.makeText(this, "Flashlight is
ON", Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Flashlight is
OFF", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```